Q1. create two different packages and demonstrate the accessibility of classes and their members using different access specifiers(public, private, default and protected).

Lab_06 > pack1 > A.java > A

```java
package Lab_06.pack1;

public class A {

    public int pub = 10;

    protected int pro = 20;

    int def = 30;           // default

    private int pri = 40;

    public void show() {

        System.out.println("Public: " + pub);

        System.out.println("Protected: " + pro);

        System.out.println("Default: " + def);

        System.out.println("Private: " + pri);

    }

}
```

Lab_06 > pack2 > B.java > B

```java
package Lab_06.pack2;

import Lab_06.pack1.A;

public class B extends A {

    Run | Debug
    public static void main(String[] args) {

        A obj = new A();
        System.out.println(obj.pub); // public

        // System.out.println(obj.pro);  not accessible
        // System.out.println(obj.def);  not accessible
        // System.out.println(obj.pri);  not accessible

        B obj2 = new B();
        System.out.println(obj2.pro); // protected via inheritance
    }
}
```

OUTPUT:—

```
gauravmathur@Annus-Mac LabJava % javac Lab_06/pack1/A.java Lab_06/pack2/B.java
gauravmathur@Annus-Mac LabJava % java Lab_06.pack2.B
10
20
gauravmathur@Annus-Mac LabJava %
```

Q2. create a package containing multiple classes and demonstrate how to access them from another package using
(i). importing all classes(package.*)
(ii). importing specific classes
(iii). using fully qualified class name

Lab_06 > mypack > A.java > A

```java
package Lab_06.mypack;
public class A {

    public void showA() {

        System.out.println(x: "This is class A");

    }

}
```

Lab_06 > mypack > B.java > B

```java
package Lab_06.mypack;

public class B {

    public void showB() {

        System.out.println(x: "This is class B");

    }

}
```

Lab_06 > Testpackage > Test1.java > Test1

```java
package Lab_06.Testpackage;

// Importing all classes from mypack package
import Lab_06.mypack.*;

public class Test1 {
    Run | Debug
    public static void main(String[] args) {

        // Creating object of class A
        A obj1 = new A();

        // Creating object of class B
        B obj2 = new B();

        // Calling methods of class A and B
        obj1.showA();
        obj2.showB();
    }
}
```

Lab_06 > Testpackage > Test2.java > Test2

```java
package Lab_06.Testpackage;

// Importing only required classes
import Lab_06.mypack.A;
import Lab_06.mypack.B;

public class Test2 {
    Run | Debug
    public static void main(String[] args) {

        // Creating object of class A
        A obj1 = new A();

        // Creating object of class B
        B obj2 = new B();

        // Calling methods
        obj1.showA();
        obj2.showB();
    }
}
```

Lab_06 > Testpackage > Test3.java > Test3

```java
package Lab_06.Testpackage;

// No import statement used here

public class Test3 {
    Run | Debug
    public static void main(String[] args) {

        // Using fully qualified names (package + class)
        Lab_06.mypack.A obj1 = new Lab_06.mypack.A();
        Lab_06.mypack.B obj2 = new Lab_06.mypack.B();

        // Calling methods
        obj1.showA();
        obj2.showB();
    }
}
```

OUTPUT:—

```
This is class B
gauravmathur@Annus-Mac LabJava % javac -d . Lab_06/mypack/*.java Lab_06/Testpackage/*.java
gauravmathur@Annus-Mac LabJava % java Lab_06.Testpackage.Test1
This is class A
This is class B
gauravmathur@Annus-Mac LabJava % java Lab_06.Testpackage.Test2
This is class A
This is class B
gauravmathur@Annus-Mac LabJava % java Lab_06.Testpackage.Test3
This is class A
This is class B
gauravmathur@Annus-Mac LabJava %
```

Q3. Write a program to demonstrate that private members are not inherited in a subclass.

Lab_06 > TestPrivate.java > Child > display()

```java
package Lab_06;

// Parent class
class Parent {

    // Private variable (cannot be accessed in subclass)
    private int x = 10;

    // Public method to access private variable
    public void showX() {
        System.out.println("Value of x: " + x);
    }
}

// Child class (inherits Parent)
class Child extends Parent {

    public void display() {

        // This will give error if uncommented
        // System.out.println(x);

        System.out.println(x: "Cannot access private variable x directly");

        // Access through public method of parent
        showX();
    }
}

// Main class
public class TestPrivate {
    Run | Debug
    public static void main(String[] args) {

        Child obj = new Child();

        // Calling child method
        obj.display();
    }
}
```

OUTPUT:—

```
gauravmathur@Annus-Mac LabJava % java Lab_06.TestPrivate
Cannot access private variable x directly
Value of x: 10
gauravmathur@Annus-Mac LabJava %
```

Q4. Write a java program consisting of a parent class and a child class, where each class contains two constructor. Demonstrate constructor chaining such that, upon object creation of the child class, all constructor of both the parent and child classes are invoked.

Lab_06 > TestConstructor.java > TestConstructor > main(String[])

```java
package Lab_06;
// Parent class
class Parent {

    // Default constructor
    Parent() {
        System.out.println(x: "Parent Default Constructor");
    }

    // Parameterized constructor
    Parent(int x) {
        System.out.println("Parent Parameterized Constructor: " + x);
    }
}
class Child extends Parent {

    // Default constructor
    Child() {
        // Calling parent default constructor
        super();
        System.out.println(x: "Child Default Constructor");
    }

    // Parameterized constructor
    Child(int x) {
        // Calling parent parameterized constructor
        super(x);
        System.out.println("Child Parameterized Constructor: " + x);
    }
}
public class TestConstructor {
    Run | Debug
    public static void main(String[] args) {

        // Creating object using default constructor
        Child obj1 = new Child();

        System.out.println(x: "------------------");

        // Creating object using parameterized constructor
        Child obj2 = new Child(x: 10);
    }
}
```

OUTPUT:—

```
gauravmathur@Annus-Mac LabJava % javac -d . Lab_06/TestConstructor.java
gauravmathur@Annus-Mac LabJava % java Lab_06.TestConstructor
Parent Default Constructor
Child Default Constructor
------------------
Parent Parameterized Constructor: 10
Child Parameterized Constructor: 10
```

Q5. Write a java Program that accepts a sentence(group of words) as command-line input. The program should count the total number of characters in all the words excluding spaces.

A.class | Test1.java U | Test2.java U | Test3.java U | TestPrivate.java U

Lab_06 > CountChars.java > CountChars

```java
package Lab_06;

// Program to count total number of characters excluding spaces
public class CountChars {
    Run | Debug
    public static void main(String[] args) {

        // Check if input is given
        if (args.length == 0) {
            System.out.println(x: "Please provide a sentence as command-line input");
            return;
        }

        int count = 0;

        // Loop through all arguments (words)
        for (String word : args) {
            count += word.length();   // count characters in each word
        }

        // Display result
        System.out.println("Total characters (excluding spaces): " + count);
    }
}
```

OUTPUT:—

```
gauravmathur@Annus-Mac LabJava % javac -d . Lab_06/CountChars.java
gauravmathur@Annus-Mac LabJava % java Lab_06.CountChars Hello World Java
Total characters (excluding spaces): 14
gauravmathur@Annus-Mac LabJava %
```